Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

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Chapter Summary

Sets

- The Language of Sets
- Set Operations
- Set Identities

Functions

- Types of Functions
- Operations on Functions
- Computability

Sequences and Summations

- Types of Sequences
- Summation Formulae

Set Cardinality

- Countable Sets

Matrices

- Matrix Arithmetic

Sets

Section 2.1

Section Summary

Definition of sets

Describing Sets

- Roster Method
- Set-Builder Notation

Some Important Sets in Mathematics

Empty Set and Universal Set

Subsets and Set Equality

Cardinality of Sets

Tuples

Cartesian Product

2.1 Sets

A *set* is an unordered collection of objects.

- the students in this class
- the chairs in this room

The objects in a set are called the *elements*, or *members* of the set. A set is said to *contain* its elements.

The notation $a \in A$ denotes that a is an element of the set A.

If a is not a member of A, write $a \notin A$

Describing a Set: Roster Method

$S = \{a,b,c,d\}$

Order not important

$S = \{a,b,c,d\} = \{b,c,a,d\}$

Each distinct object is either a member or not; listing more than once does not change the set.

$S = \{a,b,c,d\} = \{a,b,c,b,c,d\}$

Ellipses (…) may be used to describe a set without listing all of the members when the pattern is clear.

$S = \{a,b,c,d, \ldots\ldots,z\}$

Roster Method

Set of all vowels in the English alphabet:

$V = \{a,e,i,o,u\}$

Set of all odd positive integers less than 10:

$O = \{1,3,5,7,9\}$

Set of all positive integers less than 100:

$S = \{1,2,3,\ldots\ldots,99\}$

Set of all integers less than 0:

$S = \{\ldots, -3,-2,-1\}$

Some Important Sets

N = *natural numbers* = {0,1,2,3….}

Z = *integers* = {…,-3,-2,-1,0,1,2,3,…}

$\mathbf{Z}^+$ = *positive integers* = {1,2,3,…..}

R = set of *real numbers*

$\mathbf{R}^+$ = set of *positive real numbers*

C = set of *complex numbers*.

Q = set of rational numbers

Set-Builder Notation

Specify the property or properties that all members must satisfy:

$S = \{x \mid x \text{ is a positive integer less than } 100\}$

$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$

$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$

A predicate may be used: $S = \{x \mid P(x)\}$

Example: $S = \{x \mid \text{Prime}(x)\}$

Positive rational numbers:

$\mathbf{Q}^+ = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p,q\}$

Interval Notation

$a, b \in \mathbf{R}$, and $a < b$ then

- $(a, b) = \{x \in \mathbf{R} \mid a < x < b\}$
- $[a, b] = \{x \in \mathbf{R} \mid a \leq x \leq b\}$
- $(a, b] = \{x \in \mathbf{R} \mid a < x \leq b\}$
- $(-\infty, b] = \{x \in \mathbf{R} \mid x \leq b\}$
- $[a, \infty) = \{x \in \mathbf{R} \mid a \leq x\}$
- $(a, \infty) = \{x \in \mathbf{R} \mid a < x\}$

closed interval [a,b]

open interval (a,b)

Universal Set and Empty Set

The *universal set U* is the set containing everything currently under consideration.

- Sometimes implicit
- Sometimes explicitly stated.
- Contents depend on the context.

The empty set (null set) is the set with no elements.

- $\emptyset$
- $\{\}$

Some things to remember

Sets can be elements of sets.

$\{\{1,2,3\}, a, \{b,c\}\}$

$\{\mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R}\}$

A set with one element is called a singleton set.

The empty set is different from a set containing the empty set.

$\emptyset \neq \{ \emptyset \}$

Set Equality

Definition: Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \Leftrightarrow x \in B)$
- We write $A = B$ if A and B are equal sets.

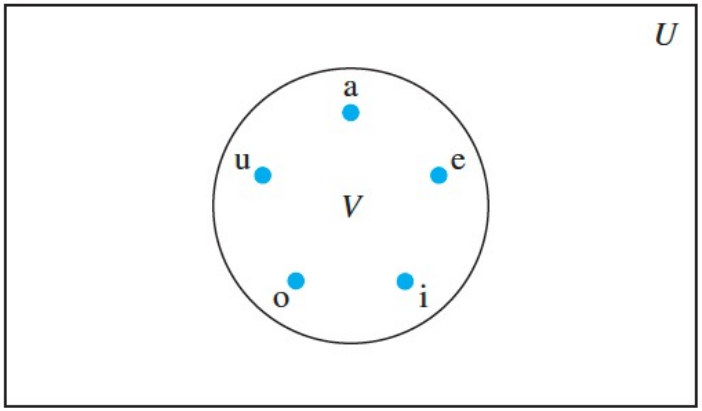
$\{1,3,5\} = \{3, 5, 1\}$

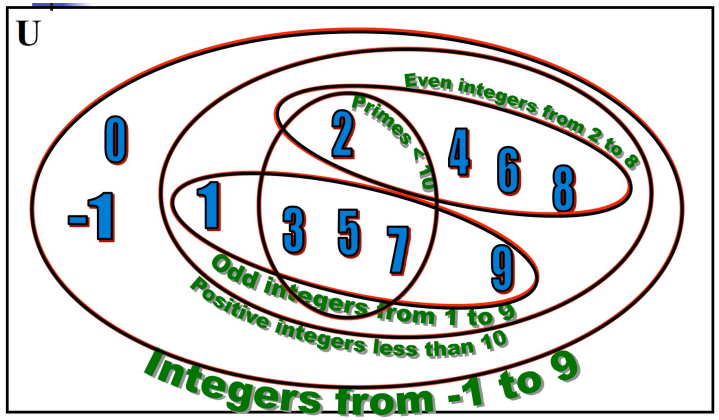
$\{1,5,5,5,3,3,1\} = \{1,3,5\}$

2.1.2 Venn Diagrams

John Venn (1834-1923)
Cambridge, UK

- In Venn diagrams the universal set U, which contains all the objects under consideration, is represented by a rectangle.
- Inside this rectangle, circles or other geometrical figures are used to represent sets.
- Sometimes points are used to represent the particular elements of the set.
- Venn diagrams are often used to indicate the relationships between sets.





2.1.3 Subsets

Definition: The set *A* is a *subset* of *B*, if and only if every element of *A* is also an element of *B*.

- The notation $A \subseteq B$ (or $B \supseteq A$) is used to indicate that *A* is a subset of the set *B*.
- $A \subseteq B$ holds if and only if $\forall x(x \in A \rightarrow x \in B)$ is true.
- Because $a \in \emptyset$ is always false, $\emptyset \subseteq S$,for every set S.
- Because $a \in S \rightarrow a \in S$, $S \subseteq S$, for every set S.

Showing a Set is or is not a Subset of Another Set

Showing that A is a Subset of B:

To show that $A \subseteq B$, show that if x belongs to A, then x also belongs to B.

Showing that A is not a Subset of B:

To show that A is not a subset of B, $A \nsubseteq B$**,** find an element $x \in A$ with $x \notin B$**.** (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Examples:

1. The set of all computer science majors at your school is a subset of all students at your school.
2. The set of integers with squares less than 100 is not a subset of the set of nonnegative integers.

Another look at Equality of Sets

Recall that two sets *A* and *B* are *equal*, denoted by *A* = *B*, iff

$$\forall x(x \in A \leftrightarrow x \in B)$$

Using logical equivalences we have that *A* = *B* iff

$$\forall x[(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

This is equivalent to

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

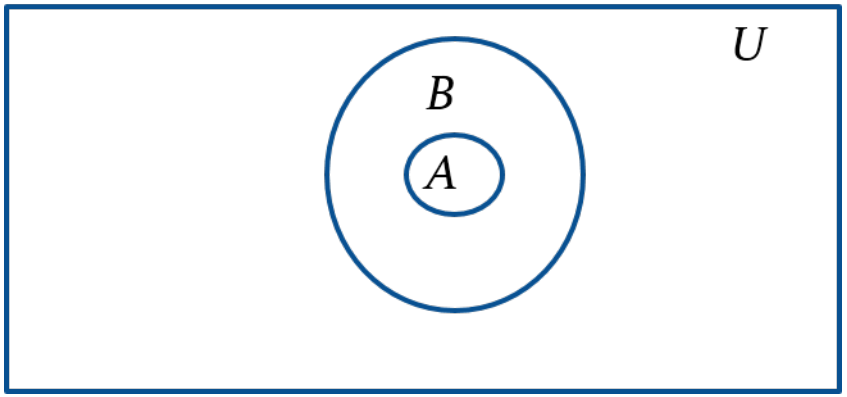
Proper Subsets

Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a *proper subset* of B, denoted by $A \subset B$. If $A \subset B$, then

$$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

is true.

Venn Diagram



2.1.4 The Size of a Set

Definition: If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition: The *cardinality* of a finite set A, denoted by $|A|$, is the number of (distinct) elements of A.

Examples:

1. $|\emptyset| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

2.1.5 Power Sets

Definition: The set of all subsets of a set *A*, denoted $\mathcal{P}$**(***A***)**, is called the *power set* of *A*.

Example: If *A* = {a,b} then

$$\mathcal{P}(\mathrm{A}) = \{\emptyset, \{a\},\{b\},\{a,b\}\}$$

If a set has *n* elements, then the cardinality of the power set is 2^n. (In Chapters 5 and 6, we will discuss different ways to show this.)

2.1.5 Power Sets

Examples

- $\mathcal{P}(\{a, b\}) =$

  $\{ \emptyset, \{a\}, \{b\}, \{a, b\} \}$

- $\mathcal{P}(\{0, 1, 2\}) =$

  $\{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\},\{0, 1, 2\}\}$

- $\mathcal{P}(\emptyset) =$

  $\{\emptyset\}$

- $\mathcal{P}(\{\emptyset\}) =$

  $\{\emptyset, \{\emptyset\}\}$

2.1.6 Cartesian Product: Tuples

The *ordered n-tuple* $(a_1,a_2,\ldots\ldots,a_n)$ is the ordered collection that has a_1 as its first element and a_2 as its second element and so on until a_n as its last element.

We say that two ordered n-tuples are equal if and only if each corresponding pair of their elements is equal.

In other words, $(a_1, a_2,\ldots, a_n) = (b_1, b_2,\ldots, b_n)$ if and only if $a_i = b_i$, for $i = 1, 2,\ldots, n$.

2.1.6 Cartesian Product: Tuples

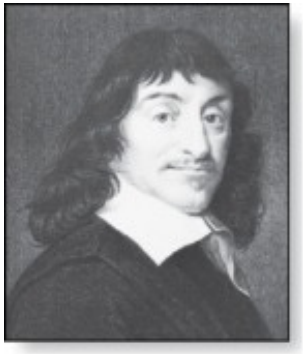
In particular, ordered 2-tuples are equal if and only if their corresponding elements are equal.

2-tuples are called *ordered pairs*.

The ordered pairs (a,b) and (c,d) are equal if and only if $a = c$ and $b = d$.

2.1.6 Cartesian Product[1]

René Descartes
(1596-1650)



Definition: The *Cartesian Product* of two sets A and B, denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a,b) | a \in A \wedge b \in B\}$$

Example:

$A = \{a,b\}$ $B = \{1,2,3\}$

$A \times B = \{(a,1),(a,2),(a,3), (b,1),(b,2),(b,3)\}$

Definition: The subset R of the Cartesian product $A \times B$ is called a *relation* from the set A to the set B. (Relations will be covered in depth in Chapter 9.)

Cartesian Product 2

Definition: The Cartesian products of the sets $A_1, A_2, \ldots\ldots, A_n$, denoted by $A_1 \times A_2 \times \ldots\ldots \times A_n$, is the set of ordered n-tuples $(a_1, a_2, \ldots\ldots, a_n)$ where a_i belongs to A_i for $i = 1, \ldots n$.

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2 \cdots, a_n) | a_i \in A_i \text{ for } i = 1,2, \ldots n\}$$

Example: What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution:

$A \times B \times C$ = {(0,1,0), (0,1,1), (0,1,2),(0,2,0), (0,2,1), (0,2,2),(1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)}

2.1.8 Truth Sets of Quantifiers

Given a predicate *P* and a domain *D*, we define the *truth set* of *P* to be the set of elements in *D* for which *P*(*x*) is true. The truth set of *P*(x) is denoted by

$$\{x \in D | P(x)\}$$

What are the truth sets of the predicates P(x), Q(x), and R(x), where the domain is the set of integers and P(x) is "|x| = 1," Q(x) is "$x^2 = 2$," and R(x) is "|x| = x."

P is the set {−1, 1}.
Q is the ø.
R is N (the set of nonnegative integers).

2.1.8 Truth Sets of Quantifiers

$\forall x P(x)$ is true over the domain U if and only if the truth set of P is the set U.

$\exists x P(x)$ is true over the domain U if and only if the truth set of P is nonempty.

Homework after class

2.1 Sets

- 22
- 23
- 25
- 47

Set Operations

Section 2.2

Section Summary

Set Operations

- Union
- Intersection
- Difference
- Complementation

More on Set Cardinality

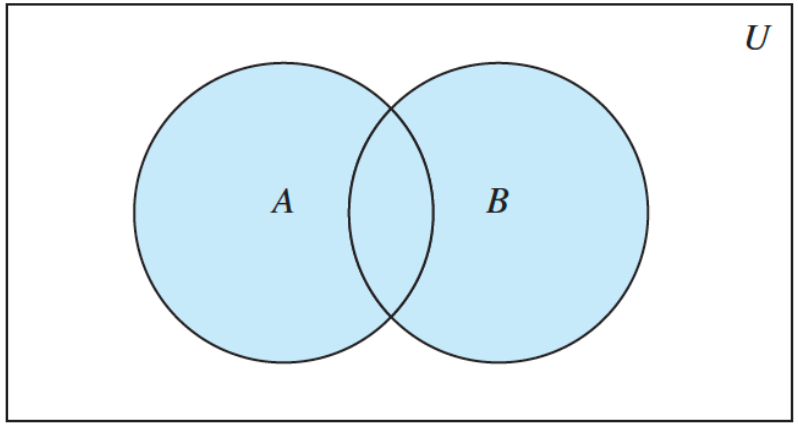
Set Identities

Proving Identities

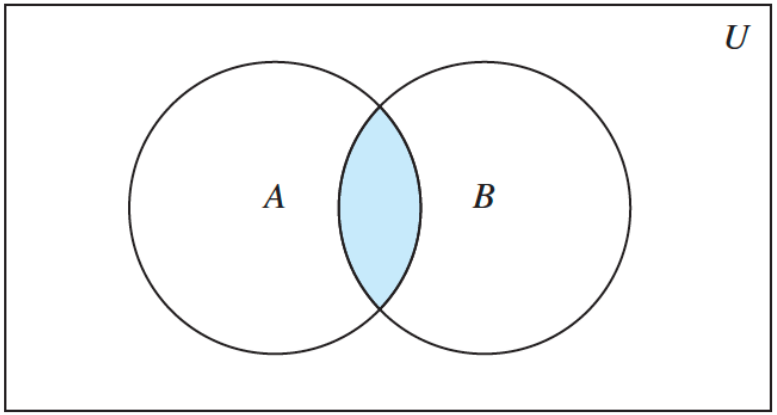
Membership Tables

2.2.1 Introduction

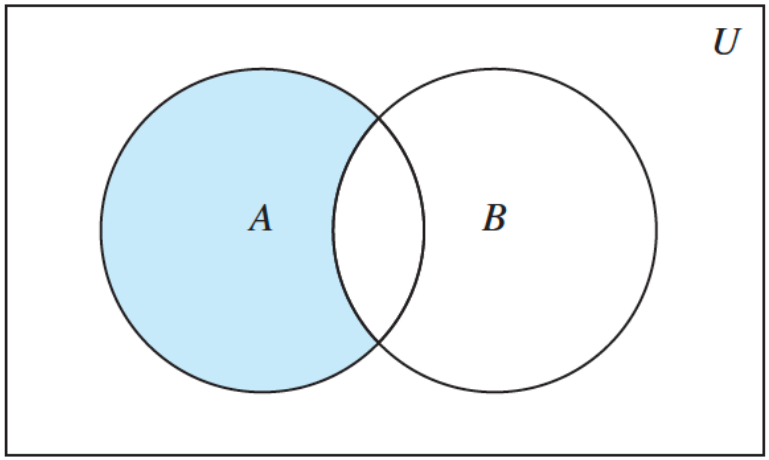
Set Operations



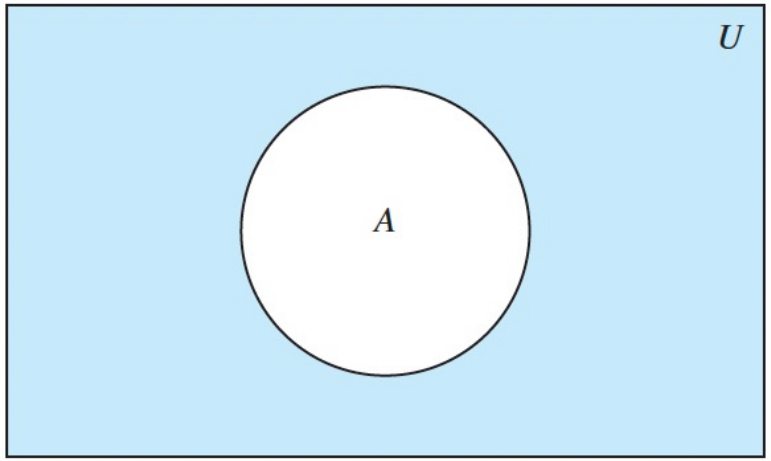
Union $A \cup B$ is shaded.



Intersection $A \cap B$ is shaded.



Difference $A - B$ is shaded.

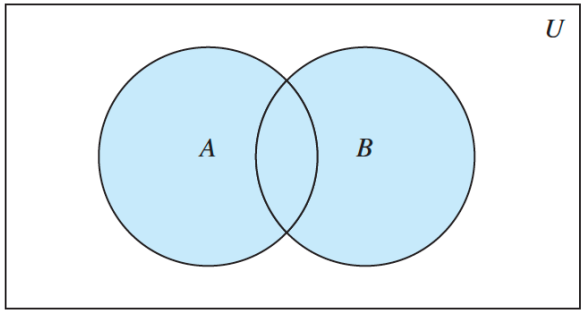


Complementation $\overline{A}$ is shaded.

Union

Definition: Let *A* and *B* be sets. The *union* of the sets *A* and *B*, denoted by $A \cup B$, is the set:

$$\{x | x \in A \vee x \in B\}$$

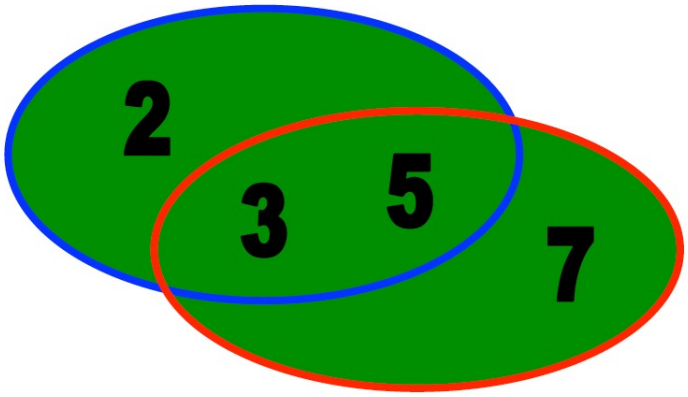


$A \cup B$ is shaded.

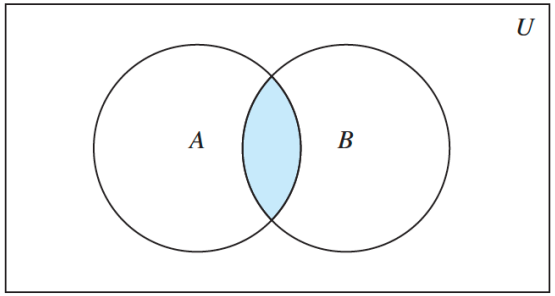
Example: What is {2,3,5} ∪ {3, 5,7}?

Solution:

{2,3,5,7}



Intersection



$A \cap B$ is shaded.

Definition: The *intersection* of sets *A* and *B*, denoted by $A \cap B$, is

$$\{x \mid x \in A \wedge x \in B\}$$

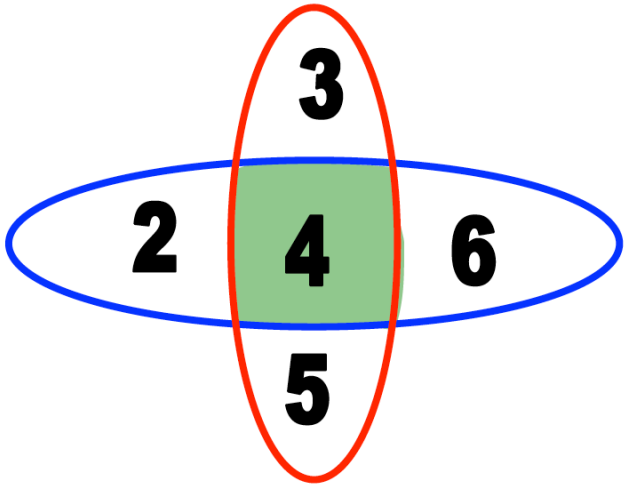
Note if the intersection is empty, then *A* and *B* are said to be *disjoint*.

Example: {2,4,6} ∩ {3,4,5} ?

Solution: {4}

Example: {1,2,3} ∩ {4,5,6} ?

Solution: ∅



Difference

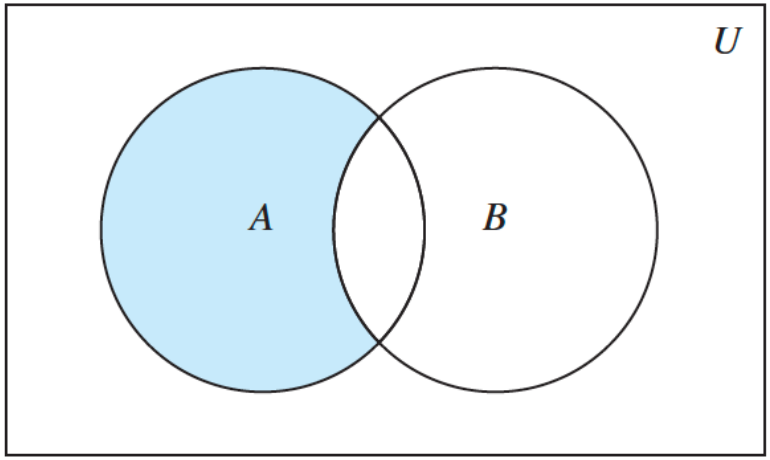
Definition: Let *A* and *B* be sets. The *difference* of *A* and *B*, denoted by *A – B (or A\B)*, is the set containing the elements of *A* that are not in *B*. The difference of *A* and *B* is also called the complement of *B* with respect to *A*.

$$A - B = \{x | x \in A \wedge x \notin B\} = A \cap \overline{B}$$

Example: {1,2,3} - {3, 4, 5}?

Solution:

{1,2}



$A - B$ is shaded.

Complement

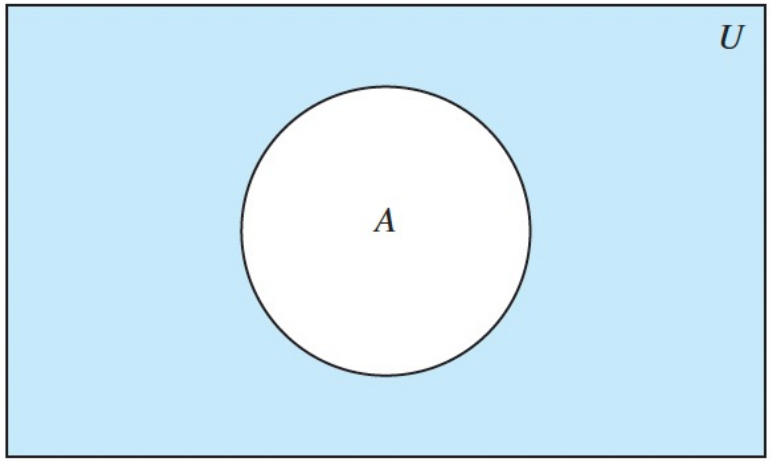
Definition: If *A* is a set, then the *complement* of the *A* (with respect to *U*), denoted by $\bar{A}$ is the set *U* - *A*

$$\bar{A} = \{x | x \in U | x \notin A\}$$

$$\overline{A} = \{x \in U \mid x \notin A\}$$

Example: If *U* is the positive integers less than 100, what is the complement of $\{x|x > 70\}$

Solution: $\{x|x \leq 70\}$

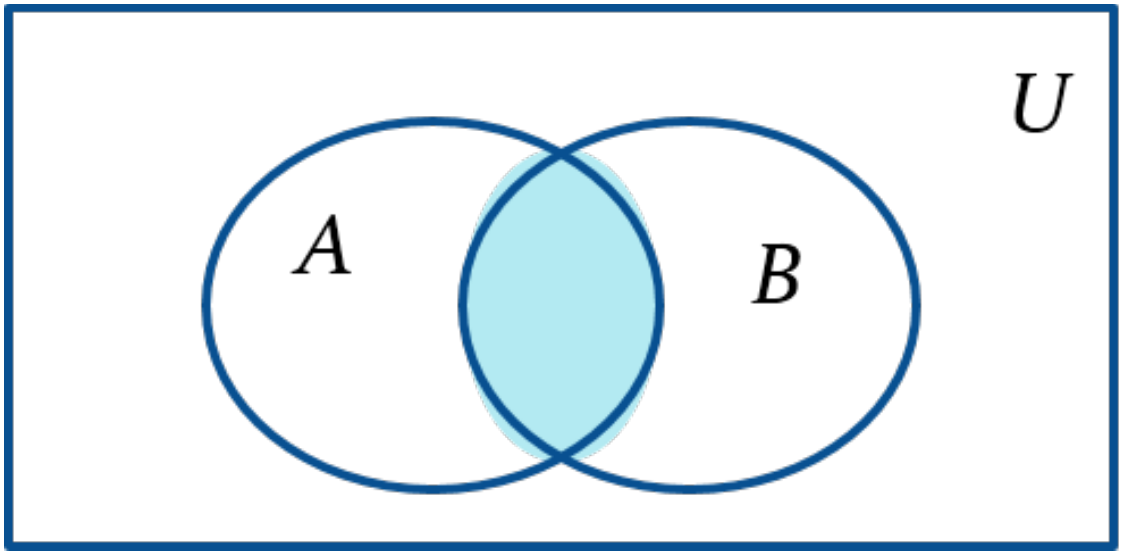


$\overline{A}$ is shaded.

The Cardinality of the Union of Two Sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Venn Diagram for A, B, A ∩ B, A ∪ B



Review Questions

$U = \{0,1,2,3,4,5,6,7,8,9,10\}$

$A = \{1,2,3,4,5\}, \quad B = \{4,5,6,7,8\}$

- $A \cup B$ $\{1,2,3,4,5,6,7,8\}$
- $A \cap B$ $\{4,5\}$
- $\bar{A}$ $\{0,6,7,8,9,10\}$
- $\bar{B}$ $\{0,1,2,3,9,10\}$
- $A - B$ $\{1,2,3\}$
- $B - A$ $\{6,7,8\}$

Symmetric Difference (*optional*)

Definition: The *symmetric difference* of **A** and **B**, denoted by $A \oplus B$ is the set

$$(A - B) \cup (B - A)$$

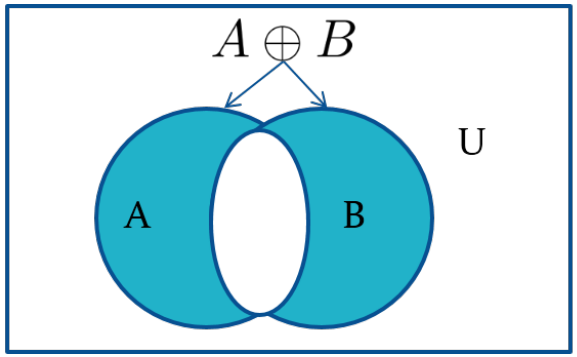
Example:

U = {0,1,2,3,4,5,6,7,8,9,10}

A = {1,2,3,4,5} B ={4,5,6,7,8}

What is $A \oplus B$:

Solution: {1,2,3,6,7,8}



Venn Diagram

2.2.2 Set Identities

- Identity: $A \cup \varnothing = A = A \cap U$
- Domination: $A \cup U = U$, $A \cap \varnothing = \varnothing$
- Idempotent: $A \cup A = A$, $A \cap A = A$
- Double complement: $\overline{(\overline{A})} = A$
- Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C$,
  $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,
  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- Absorption: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$
- Complement: $A \cup \overline{A} = U$, $A \cap \overline{A} = \varnothing$
- De Morgan's $\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Proving Set Identities

Different ways to prove set identities:

- Subset method: Show that each side of the identity is a subset of the other side.
- Membership table: For each possible combination of the atomic sets, show that an element in exactly these atomic sets must either belong to both sides or belong to neither side.
- Apply existing identities: Start with one side, transform it into the other side using a sequence of steps by applying an established identity.
- Venn Diagram

Way 1 : Subset method 1

Example: Prove that De Morgan Law $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution: We prove this identity by showing that:

1) $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$ and
2) $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

Way 1 : Subset method 2

These steps (1) show that: $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

$x \in \overline{A \cap B}$ by assumption

$x \notin A \cap B$ defn. of complement

$\neg\big((x \in A) \wedge (x \in B)\big)$ by defn. of intersection

$\neg(x \in A) \vee \neg(x \in B)$ 1st De Morgan law for Prop Logic

$x \notin A \vee x \notin B$ defn. of negation

$x \in \overline{A} \vee x \in \overline{B}$ defn. of complement

$x \in \overline{A} \cup \overline{B}$ by defn. of union

Way 1 : Subset method 3

These steps (2) show that: $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

$x \in \overline{A} \cup \overline{B}$	by assumption
$\left(x \in \overline{A}\right) \vee \left(x \in \overline{B}\right)$	by defn. of union
$\left(x \notin A\right) \vee \left(x \in \overline{B}\right)$	defn. of complement
$\neg\left(x \in A\right) \vee \neg\left(x \in B\right)$	defn. of negation
$\neg\left(\left(x \in A\right) \wedge \left(x \in B\right)\right)$	1st De Morgan law for Prop Logic
$\neg\left(x \in A \cap B\right)$	defn. of intersection
$x \in \overline{A \cap B}$	defn. of complement

Way 2 : Set-Builder Notation

$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}.$$

$\overline{A \cap B} = \{x \mid x \notin A \cap B\}$	by definition of complement
$= \{x \mid \neg(x \in (A \cap B))\}$	by definition of does not belong symbol
$= \{x \mid \neg(x \in A \wedge x \in B)\}$	by definition of intersection
$= \{x \mid \neg(x \in A) \vee \neg(x \in B)\}$	by the first De Morgan law for logical equivalences
$= \{x \mid x \notin A \vee x \notin B\}$	by definition of does not belong symbol
$= \{x \mid x \in \overline{A} \vee x \in \overline{B}\}$	by definition of complement
$= \{x \mid x \in \overline{A} \cup \overline{B}\}$	by definition of union
$= \overline{A} \cup \overline{B}$	by meaning of set builder notation

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

A	***B***	***C***	***B*∩*C***	***A*∪ (*B*∩*C*)**	***A*∪*B***	***A*∪*C***	***(A*∪*B)* ∩ *(A*∪*C)***

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1	1	1					
1	1	0					
1	0	1					
1	0	0					
0	1	1					
0	1	0					
0	0	1					
0	0	0					

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Solution:

A	***B***	***C***	***B∩C***	***A∪ (B∩C)***	***A∪B***	***A∪C***	***(A∪B) ∩ (A∪C)***
1	1	1	1				
1	1	0	0				
1	0	1	0				
1	0	0	0				
0	1	1	1				
0	1	0	0				
0	0	1	0				
0	0	0	0				

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1	1	1	1	1			
1	1	0	0	1			
1	0	1	0	1			
1	0	0	0	1			
0	1	1	1	1			
0	1	0	0	0			
0	0	1	0	0			
0	0	0	0	0			

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1	1	1	1	1	1		
1	1	0	0	1	1		
1	0	1	0	1	1		
1	0	0	0	1	1		
0	1	1	1	1	1		
0	1	0	0	0	1		
0	0	1	0	0	0		
0	0	0	0	0	0		

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

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1	1	1	1	1	1	1	
1	1	0	0	1	1	1	
1	0	1	0	1	1	1	
1	0	0	0	1	1	1	
0	1	1	1	1	1	1	
0	1	0	0	0	1	0	
0	0	1	0	0	0	1	
0	0	0	0	0	0	0	

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

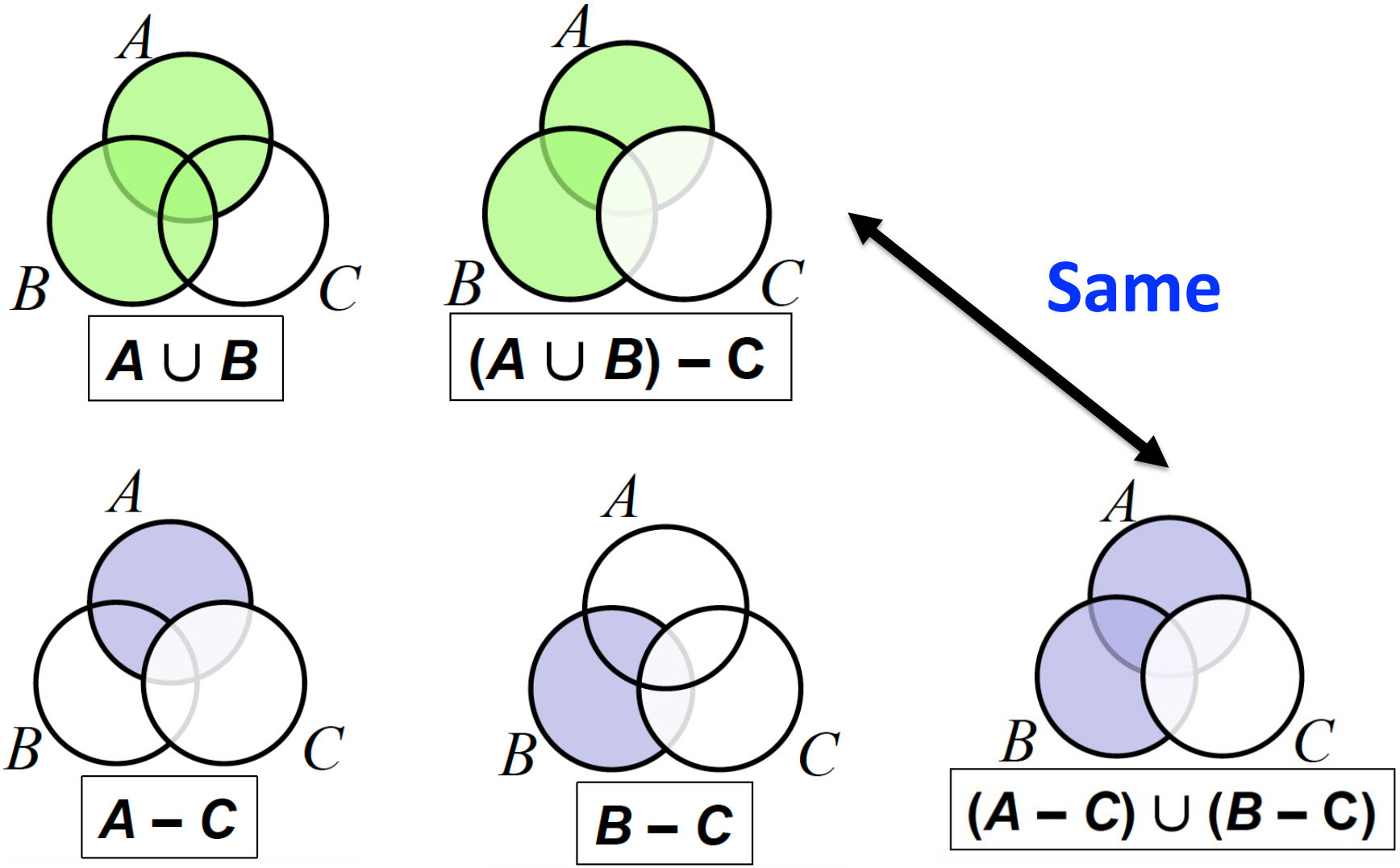
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

A	***B***	***C***	***B∩C***	***A∪ (B∩C)***	***A∪B***	***A∪C***	***(A∪B) ∩ (A∪C)***
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Way 4 : Venn Diagram

Prove (A ∪ B) - C = (A - C) ∪ (B - C).



Methods of Proving Set Identities

TABLE 3 Methods of Proving Set Identities.

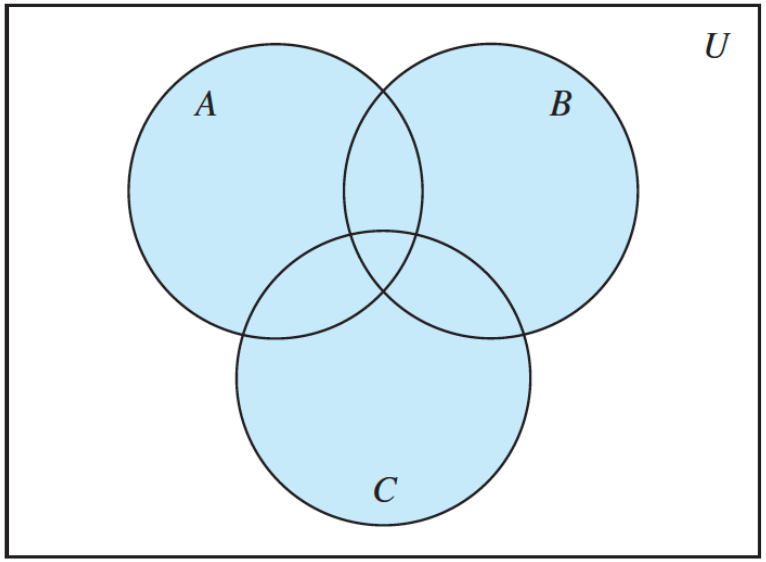
Description	*Method*
Subset method	Show that each side of the identity is a subset of the other side.
Membership table	For each possible combination of the atomic sets, show that an element in exactly these atomic sets must either belong to both sides or belong to neither side
Apply existing identities	Start with one side, transform it into the other side using a sequence of steps by applying an established identity.

Venn Diagram

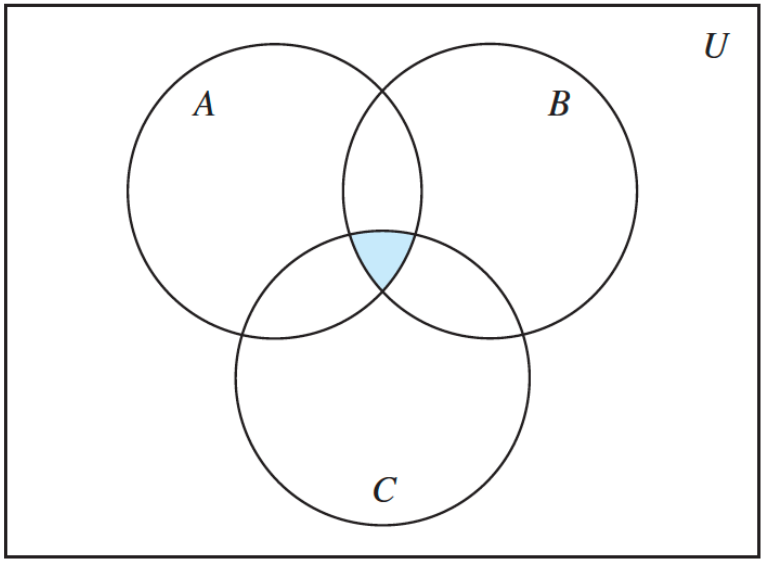
2.2.3 Generalized Unions and Intersections

Because unions and intersections of sets satisfy associative laws, the sets A ∪ B ∪ C and A ∩ B ∩ C are well defined (unambiguous);

That is, we do not have to use parentheses to indicate which operation comes first because A ∪ (B ∪ C) = (A ∪ B) ∪ C and A ∩ (B ∩ C) = (A ∩ B) ∩ C.



(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

2.2.3 Generalized Unions and Intersections

Let $A_1, A_2, \ldots, A_n$ be an indexed collection of sets.

We define:

$$\boxed{A_1 \cup A_2 \cup \cdots \cup A_n = \bigcup_{i=1}^{n} A_i}$$

$$\boxed{A_1 \cap A_2 \cap \cdots \cap A_n = \bigcap_{i=1}^{n} A_i}$$

Example: For $i = 1,2,\ldots$, let $A_i = \{i, i+1, i+2, \ldots\}$. Then,

$$\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} \{i, i+1, i+2, \ldots\} = \{1,2,3, \ldots\}$$

$$\bigcap_{i=1}^{n} A_i = \bigcap_{i=1}^{n} \{i, i+1, i+2, \ldots\} = \{n, n+1, n+2, \ldots\} = A_n$$

2.2.4 Computer Representation of Sets

- For an enumerable universal set U with ordering x_1, x_2, x_3,…, we can represent a finite set $S \subseteq U$ as the finite bit string $B = b_1b_2\ldots b_n$ where $b_i = 1$ if $x_i \in S$ and $b_i = 0$ if $x_i \notin S$.
- Example. U={1,2,3,4,5,6,7,8,9,10}
  - The subset of all even integers in U, namely, {2, 4, 6, 8, 10}, by the string: 01 0101 0101.
  - The set of all integers in U that do not exceed 5, namely, {1, 2, 3, 4, 5}, is represented by the string: 11 1110 0000.
- In this representation, the set operators "∪", "∩", "— " are implemented directly by bitwise OR, AND, NOT!

2.2.4 Computer Representation of Sets

Let U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, and the ordering of elements of U has the elements in increasing order, then

$S1 = \{1, 2, 3, 4, 5\} \Rightarrow B1 = 11\ 1110\ 0000$

$S2 = \{1, 3, 5, 7, 9\} \Rightarrow B2 = 10\ 1010\ 1010$

- $S1 \cup S2 = \{1, 2, 3, 4, 5, 7, 9\} \Rightarrow$ bit string =

  $11\ 1110\ 1010 = B1 \vee B2$

- $S1 \cap S2 = \{1, 3, 5\} \Rightarrow$ bit string =

  $10\ 1010\ 0000 = B1 \wedge B2$

- $\overline{S_1} = \{6, 7, 8, 9, 10\} \Rightarrow$ bit string =

  $00\ 0001\ 1111 = \neg B1$

2.2.5 Multisets

Sometimes the number of times that an element occurs in an unordered collection matters.

A multiset (short for multiple-membership set) is an unordered collection of elements where an element can occur as a member more than once.

We can use the same notation for a multiset as we do for a set, but each element is listed the number of times it occurs.

The multiset denoted by {a, a, a, b, b} is the multiset that contains the element a thrice and the element b twice.

2.2.5 Multisets

We can avoid this ambiguity by using an alternate notation for multisets. The notation $\{m_1 \cdot a_1, m_2 \cdot a_2, \ldots, m_r \cdot a_r\}$ denotes the multiset with element a_1 occurring m_1 times, element a_2 occurring m_2 times, and so on.

The numbers m_i, $i = 1, 2, \ldots, r$, are called the multiplicities of the elements a_i, $i = 1, 2, \ldots, r$.

Elements not in a multiset are assigned 0 as their multiplicity in this set.

The cardinality of a multiset is defined to be the sum of the multiplicities of its elements.

2.2.5 Multisets

Suppose that P and Q are the multisets $\{4 \cdot a, 1 \cdot b, 3 \cdot c\}$ and $\{3 \cdot a, 4 \cdot b, 2 \cdot d\}$, respectively. Find $P \cup Q$, $P \cap Q$, $P - Q$, and $P + Q$.

$$\begin{aligned} P \cup Q &= \{\max(4, 3) \cdot a, \max(1, 4) \cdot b, \max(3, 0) \cdot c, \max(0, 2) \cdot d\} \\ &= \{4 \cdot a, 4 \cdot b, 3 \cdot c, 2 \cdot d\}, \end{aligned}$$

$$\begin{aligned} P \cap Q &= \{\min(4, 3) \cdot a, \min(1, 4) \cdot b, \min(3, 0) \cdot c, \min(0, 2) \cdot d\} \\ &= \{3 \cdot a, 1 \cdot b, 0 \cdot c, 0 \cdot d\} = \{3 \cdot a, 1 \cdot b\}, \end{aligned}$$

$$\begin{aligned} P - Q &= \{\max(4 - 3, 0) \cdot a, \max(1 - 4, 0) \cdot b, \max(3 - 0, 0) \cdot c, \max(0 - 2, 0) \cdot d\} \\ &= \{1 \cdot a, 0 \cdot b, 3 \cdot c, 0 \cdot d\} = \{1 \cdot a, 3 \cdot c\}, \text{ and} \end{aligned}$$

$$\begin{aligned} P + Q &= \{(4 + 3) \cdot a, (1 + 4) \cdot b, (3 + 0) \cdot c, (0 + 2) \cdot d\} \\ &= \{7 \cdot a, 5 \cdot b, 3 \cdot c, 2 \cdot d\}. \end{aligned}$$

Section Summary

Set Operations

- Union
- Intersection
- Difference
- Complementation

More on Set Cardinality

Set Identities

Proving Identities

Membership Tables

Homework after class

2.2 Set Operations

- 20 (a, and c)
- 27
- 63